COMMUNICATIONS SECTOR DESIGN BASIS FRAMEWORK

Design Basis, Design Basis Accident, and Military-Action Extension Methodology
Function-Primary Facility-Class Architecture

DBA-COMM: Communications Sector Parent Document

A standalone sector series in the Design Basis for AI Governance in Critical Infrastructure

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For review by engineers, regulators, network operators, and policy analysts responsible for protecting communications-sector critical infrastructure across transport, access, core, and operations functions.

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

Section 1 — Purpose, Scope, and Series Architecture

1.1 Purpose

This document establishes the Design Basis Framework for the National Infrastructure Protection Plan (NIPP) Communications sector. It is the parent instrument for the DBA-COMM series and governs the facility-class instruments of every type developed beneath it. It applies the DB → DBA → DBA-MA three-layer analytical progression: a baseline Design Basis (DB) establishing the postulated conditions the sector's facilities must withstand; a Design Basis Accident (DBA) layer bounding the non-adversarial consequence envelope; and a Military-Action extension (DBA-MA) applying the four characteristics of deliberate adversarial action — intentionality, persistence, precision, and denial — to the sector's governing postulated events.

1.2 What This Document Does Not Do

This document does not supersede or modify any existing communications-sector regulation or standard. It does not restate the governing definitions of the Bright Line or the Command Broker, which are inherited by reference from FD-BL and its determination annexes. It does not, at this version, develop the per-class analytical content: the facility-class instruments of every type named in the register carry that work.

1.3 Function-Primary Organizing Principle

Design-basis placement follows the consequence chain a facility governs — the initiating events it must withstand and the mitigations available to answer them — not the administrative label. The Communications sector is organized here by function: the movement of information from origin to destination, decomposed into transport, access, and the switching-and-routing core, together with the operations-and-coordination function that governs them. Hazard and asset characterization are anchored to the reliability instrumentation the sector already maintains — the FCC Network Outage Reporting System (NORS) and Disaster Information Reporting System (DIRS), the CSRIC best-practice corpus, and the NSEP priority framework (GETS and WPS) — rather than a bespoke taxonomy.

1.4 Series Placement — DBA-COMM as a Standalone Sector Series

DBA-COMM is a standalone sector series, a peer of DBA-EN (Energy), DBA-W (Water), DBA-CHEM (Chemical), and the DBA-MA nuclear series. It is not subordinate to any other sector. As with every sector series, it inherits the DB → DBA → DBA-MA methodology from the DBA-MA parent and the Bright Line and Command-Broker architecture from FD-BL, and it applies the mandatory Military-Action Extension Applicability Determination. The Communications sector and the Information Technology sector together constitute the digital substrate on which the other sectors operate; DBA-COMM addresses the transport-and-connectivity substrate, and the DBA-DC (data center) series addresses the compute substrate.

1.5 Relationship to DBA-MA-CDB — the Supply / Demand Dual

The corpus already contains a communications design basis: DBA-MA-CDB (*Military Action Communications Design Basis*), held at the DBA-MA parent, which sets out seven requirements, CDB-1 through CDB-7. It is essential to state precisely what that instrument is and is not, because the distinction defines this framework. It is the demand-side instrument: it specifies how a facility in any sector must be designed to operate when an adversary degrades or denies the communications it depends on — degradation as a baseline condition, local protective autonomy, field-executable procedures, verified independent channels, coordination without shared connectivity, Command-Broker continuity under denial, and regular testing of the independent channels. It does not describe the communications infrastructure itself.

DBA-COMM is the supply-side dual. It establishes the design basis for the communications infrastructure whose loss produces exactly the condition every other sector must design to. Seen from any dependent sector, the failure of a DBA-COMM facility class is the onset of the degraded baseline that CDB-1 describes. A communications facility is also a dependent in its own right: its network management, its alarms and its operators' instructions travel over communications links, and it must keep working when those links are degraded or denied. For those links the *Military Action Communications Design Basis* applies to a communications facility exactly as it applies to a facility in any other sector, under its own §3. This framework therefore cites that document and does not restate or adapt it. Each DBA-COMM facility-class instrument states how the seven requirements apply to its own control and coordination communications, as *Military Action Communications Design Basis* §3 requires of every facility class, and records any justified exception in the way that section directs.

Section 2 — The Communications Sector and Its Regulatory Context

2.1 A Hybrid, Largely-Private, Lightly-Mandated Sector

The Communications sector is privately owned and operated, coordinated through a hybrid of regulation and voluntary partnership rather than a single mandatory reliability regime. The Federal Communications Commission holds rulemaking and outage-reporting authority (NORS, DIRS). The Cybersecurity and Infrastructure Security Agency is the Sector Risk Management Agency and operates the National Coordinating Center for Communications and the sector's priority services (GETS, WPS). Reliability practice is developed largely through the voluntary Communications Security, Reliability and Interoperability Council (CSRIC) and the President's National Security Telecommunications Advisory Committee (NSTAC). No mandatory engineering design basis of the kind the nuclear and electric sectors carry exists for communications infrastructure.

2.2 Coordination-Per-Function Inheritance

As in the other sectors, the applicable authority varies by function, so the function-primary taxonomy is also the natural structure for the regulatory and coordination terrain. Transport carries FCC infrastructure rules and, for subsea cable and foreign-carrier involvement, Team Telecom review; access carries FCC and state authority and, for public-safety broadband, the FirstNet framework; the switching-and-routing core carries FCC rules and the voluntary practice of the routing and signaling community; timing distribution carries the policy baseline of Executive Order 13905 and no single regulator; and the operations-and-coordination function carries the CISA sector-coordination role, the NCC, and the Communications ISAC. The domain name system and its ICANN stewardship belong to the Information Technology sector under the boundary drawn in Section 3.4.

2.3 The Regulatory Gap

No communications-sector instrument today defines an engineering design basis — the set of events the infrastructure must be demonstrated to withstand. Outage-reporting and best-practice corpora exist; a design basis does not. The gap is most acute for position, navigation, and timing: the network's dependence on GNSS timing is a single point of common-cause failure across communications, the electric grid, and financial markets, addressed at the policy level by Executive Order 13905 but carried by no facility-level design basis and governed by no single regulator.

Section 3 — Function-Primary Taxonomy

3.1 The Function Classes

The Communications sector is decomposed into five function classes. Four trace the path of information and the timing it depends on — transport, access, the switching-and-routing core, and positioning, navigation and timing — and each is realized as a physical facility class. The fifth, operations and coordination, is a coordination class. One cross-cutting overlay, critical services, rides on the physical classes and drives cross-sector elevation. Each class is placed by the consequence chain it governs and carries its own regulatory and coordination anchor. The class types follow the author's determination of 30 September 2026 (*Facility-Class Types Are Physical, Coordination and Logical*).

3.2 Transport / Backbone (TR)

The transport function covers the long-haul and metropolitan fiber, subsea cables and their landing stations, the core optical and IP backbone, and satellite backbone. Its governing consequence is loss of bulk information transport across a region or between regions, and its defining vulnerability is concentration: cable landing stations, long-haul chokepoints, and the routes adjacent to major interconnection points aggregate traffic in a way that makes a small number of physical sites disproportionately consequential. The class ends at the buildings where networks meet: on the author's decision of 6 October 2026, transport owns the fiber up to an

interconnection building or internet exchange point, and the building itself belongs to the core class (Section 3.4).

3.3 Access / Edge (AC)

The access function covers the last-mile delivery of service: the cellular radio access network and its towers and base stations, wireline and fixed-wireless local loops, satellite access, and over-the-air broadcast — the television and AM/FM and satellite radio distribution that also carries the Emergency Alert System. Two properties distinguish it. First, it is the point at which service is handed to the end user and to dependent critical facilities — the tower that serves a hospital, the circuit that serves a 911 center — so the sector's outward criticality concentrates here. Second, it is acutely power-dependent: central-office backup typically spans eight to twenty-four hours and cell-site batteries four to eight hours, making access the node where the bidirectional dependency with the electric sector is most immediate. On the author's decision of 6 October 2026, a central office building and its backup power belong to this class, and the switching and routing decisions made inside it belong to the core class (Section 3.4).

3.4 Core — Switching, Routing & Signaling (CR)

The core function covers the network's control plane: mobile switching centers, internet exchange points, the SS7 and Diameter signaling systems, BGP routing, and the number-portability databases. It is software-data dominated and is the communications analog of the electric sector's control function — the coordination brain whose compromise degrades the whole rather than a part. DNS, PKI, and identity are governed by the Information Technology sector's Connectivity & Protocols (CP) class per the ratified COMM/IT boundary; COMM-Core retains the SS7/BGP signaling and inter-carrier routing that move calls and packets between networks. It is also the locus at which the automation and AI governance concerns of Section 5 are most consequential, because routing, signaling, and traffic-management decisions increasingly involve machine-learning and generative components. Core is a physical class. Its mobile switching centers, internet exchange points and signaling nodes are sited facilities, in many ways like the data-center structures of the DBA-DC series, and the routing and signaling it governs run on equipment housed in them. The interconnection buildings and internet exchange points where networks meet belong to this class, and transport owns the fiber up to them (author's decision of 6 October 2026). Switching and routing decisions made inside a central office belong to this class as well, while the building and its backup power belong to access. The building-level protection of core sites — power, cooling, fire and physical security — is cited from the data-center series rather than restated (author's decision of 6 October 2026).

3.5 PN — PNT Reception, Timing & Synchronization Distribution

The PNT function covers the reception, holdover, and distribution of the positioning, navigation, and timing services on which the network depends — and, through the network, the finance, electric-grid, and other sectors that draw timing from the same source. It is elevated here from a

cross-cutting overlay to a function class in its own right because GPS-derived timing is a common-cause dependency that underpins communications, finance, the grid, and more, so the reception-and-distribution infrastructure warrants a design basis of its own. The class owns the ground-segment reception, the holdover that carries the network through a loss of signal, and the synchronization distribution that propagates timing across the sector. The GPS space segment — the constellation itself, operated by the United States Space Force — is a boundary object shared with the Government Facilities and Defense Industrial Base sectors and is not owned here. On the author's decision of 6 October 2026, the class also holds the design basis for timing receivers in every sector: the facility classes of other series cite it for their receivers and state, in their own terms, the consequence of timing error at their facilities.

3.6 OpsCo — Operations and Coordination (coordination class)

The operations-and-coordination function is carried by a coordination class, OpsCo, spanning the private Network Operations Centers that run each carrier and the sector-level coordination commons — the NCC, the Communications ISAC, and mutual-aid arrangements. Communications sits between the electric and chemical sectors on the coordination spectrum: unlike the chemical sector, which has no sector-level coordinator and collapses coordination into the owning firm, communications has a genuine external coordination commons; unlike the electric sector, that commons is voluntary and partnership-based rather than a mandatory reliability authority. OpsCo is the locus at which the AI-Governance Design Basis is implemented — the Bright Line placement and Command-Broker obligation for machine-learning and generative-AI systems in network operations attach here. On the author's decision of 6 October 2026, the Command Broker for an action that crosses carriers is federated: each carrier's qualified holder authorizes that carrier's part of the action, and the sector's coordination commons coordinates without holding broker authority. On the author's decision of 6 October 2026, the Command Broker's independence in this sector is procedural, as *The Broker* §4.5 allows where the deploying authority's interest is aligned with correct placement. The decisions in which a carrier's commercial interest runs against refusal — deferring a restoration that would carry a competitor's customers, declining roaming that benefits a rival, switching off coverage to save energy, delaying an outage report — are recorded in *Operations and Coordination* §6.5 rather than structurally separated. Recording them is part of the justification from the domain's incentive structure that *The Broker* §4.5 requires, and the holder's ability to refuse without answering to the person who gains remains the safeguard. Under military action the commons may not stay voluntary: in Ukraine from May 2022 the orders of the state's network-management centre bound every carrier, and from February 2026 the centre may order national roaming where no contract provides it (*Operations and Coordination* §1.3). On the author's decision of 8 October 2026, an order of a state coordinator that binds carriers is a governmental act outside the design basis, as the President's emergency powers under Section 706 of the Communications Act are, and it does not replace the carrier's holder: each carrier's qualified holder still authorizes the carrier's execution of its part in its own network, the order forming

part of the basis on which the holder acts, its integrity governed by FD-BL-D2 (*Indication Integrity*). *Operations and Coordination* §6.4 states it, and cites as practice the two features of Ukraine's Law 4670-IX: an order only where no contract provides the arrangement, and an order that names the technologies and services it requires.

† OpsCo is introduced here as a coordination-class designation; the designators below were ratified 2026-08-09. OpsCo parallels the CntlCo (electric) and CorpCo (chemical) coordination classes; the name records that communications coordination is a hybrid of the two.

3.7 Cross-Cutting Overlay: Critical Services

One overlay rides on the physical classes rather than constituting a transport path of its own. The critical-services overlay covers 911 and Next-Generation 911, the Emergency Alert System and Wireless Emergency Alerts, and the GETS and WPS priority services; it is the life-safety layer whose dependency fires the cross-sector elevation rule at the access boundary. Positioning, navigation and timing was carried here as a second overlay until it was made a facility class at v0.3 (Section 3.5). It is no longer an overlay. Its cross-sector weight — the dependence of the network, and through it of the grid and the financial markets, on satellite timing as a single point of common-cause failure — is carried by DBA-COMM-PN-FC4. On the author's decision of 6 October 2026, critical services stays an elevation overlay and does not become a facility class. The emergency services series has drawn the boundaries that decide it: 911 centers and the Next-Generation 911 network belong to DBA-ESS-AD-FC1 (*Public-Safety Answering and Dispatch*), and the origination of alerts belongs to DBA-ESS-EMCo (*Emergency Management and Coordination*), while the dissemination of alerts over broadcast and carrier networks stays with DBA-COMM-AC-FC2 (*Access and Edge*). The priority services ride on the carriers' own networks and own no facility. With those lines drawn the overlay owns no facility of its own. It names the life-safety services whose sole paths take a tightened acceptance criterion at the access boundary, *Access and Edge* §8.1 states that criterion, and no separate instrument is written for the overlay. The overlay's elevation rule counts the dependence of a 911 service on a single transport network, including paths a 911 provider obtains from a carrier that ride that carrier's one network however diverse their routes. In the CenturyLink outage of December 2018 the four diverse signaling links of a 911 processor serving Washington, north central Texas and small wireless carriers all rode the carrier's transport, and of the some seventeen million customers in twenty-nine states who lacked reliable access to 911, most lacked it through other providers' services that rode that network (*CenturyLink Outage* F-5; *Transport and Backbone* §3.2). Where a region reaches the rest of the network by one path, the surviving service carried while that path is down includes, beside the life-safety services the overlay names, the payment and settlement links on which the region's economy depends and the coordination traffic of a disaster response, and the method for setting the surviving service from a region's dependents counts them; *Transport and Backbone* takes the element in at v0.3, at its §3.4 and §10.1. While the one international cable of the Kingdom of Tonga was down after the eruption of January 2022 (DBA-COMM-RE-3), teller machines stopped because banks could not check balances, export

documents could not be filed, and remittances, which the Guardian and the Devpolicy Blog put at 44 per cent of gross domestic product, could not be sent, and the government's coordination with its emergency management office and the humanitarian coordination were hampered (*Tonga Isolation* F-5).

Section 4 — The Communications Sector Design Basis

[Skeleton — to be developed.] The design envelope will be organized by event category with a consequence-severity inclusion criterion consistent with the series.

4.1 Design Basis Principles

The criterion for inclusion in the design envelope is consequence severity — potential for loss of the communications service on which public safety and other critical sectors depend, at population or regional scale, or long-duration denial to a critical dependent facility — not event likelihood. Candidate bounding events, to be developed: backbone or landing-station severance; regional access denial; core compromise (routing or signaling); GNSS timing loss; and the mass-outage / restoration-under-degradation case.

Among the initiating events of backbone or landing-station severance is a ship disabled, abandoned or sunk by military action within reach of a cable corridor, and its wreck, which is a hazard to repair as well as an initiating event. In the Red Sea cable cuts of February 2024 (DBA-COMM-RE-2), the operators, the International Cable Protection Committee and the United States government give as the likely cause the dragged anchor of the Rubymar, a cargo ship struck by a Houthi missile six days before and abandoned by its crew, and its wreck, which Seacom estimated lay about a kilometer from its cable, was a hazard to the repair. The waters of a war against shipping are counted as a source of such hazards whether or not the cables are a target (*Red Sea Cable Cuts* F-3).

Repair capacity in a war is the part of the repair fleet that its owners will send, that insurers will cover and that the coastal authorities will admit, which may be one ship. That is the reading of *Red Sea Cable Cuts*, which this framework adopts: one contractor, E-Marine, and one ship, the Niwa, repaired all three Red Sea systems in turn. A reserve of repair capacity, if the design basis holds one, is sized against that part of the fleet and not against the fleet, and the Red Sea repair, about five months from 24 February to July 2024, is a point on the record for subsea repair in contested waters (*Red Sea Cable Cuts* F-6; *Transport and Backbone* §10.2, §10.8).

The framework adopts for the sector the consequence-basis conditionality principle its classes state (*Transport and Backbone* §5.8, §10.1; *Access and Edge* §5.8, §10.1): any relaxation of a conservative class parameter is justified against the class baseline, the larger the relaxation the stronger the analytical basis required, and a relaxation without a recorded basis is not consistent with the design basis. It takes one trigger with the principle. The loss of a sole path is the residual of an isolated region realized, and a path count of one accepted after such a loss carries a recorded basis that states the interim surviving service and its reserve and the date by which a

second path is expected. Tonga lost its sole international path in January 2019 and again in January 2022, and its northern islands lost their one cable to the capital in 2019, 2022 and 2024; a second international cable, explored in 2019, was switched on in 2026 (*Tonga Isolation* F-7; *Transport and Backbone* §5.7, §5.8).

Section 5 — Dual Design Basis: Bright Line and Command Broker

[Skeleton — to be developed on the DBA-W §5.5 model.]

Any communications facility may simultaneously carry a physical Design Basis and an AI-Governance Design Basis governing machine-learning and generative-AI systems in its operations. The core (CR) and operations (OpsCo) functions are the primary loci: the actions of in-scope components in autonomous routing and traffic management and in self-organizing-network optimization are those whose Bright Line placement must be established, each on its own consequence, with the Command Broker required for any above-line action that could itself precipitate a regional outage. Generative decision-support output in the NOC that would make no change of its own is placed below the line as a process action and carries the indication-integrity obligation of FD-BL-D2 wherever it can ground an above-line authorization (FD-BL §4.2); a change it recommends is one action with its acceptance and is placed on that change's consequence (FD-BL §4.3). Determinism decides scope under FD-LD §4.3; it does not decide placement. Consistent with CDB-6, communications degradation does not suspend the Command-Broker obligation.

Section 6 — Cross-Sector Dependencies and the Consequence Elevation Rule

6.1 Outward View — Communications Failure as the CDB Baseline

The Communications sector is the widest foundational dependency in the critical-infrastructure framework. Its outward consequence is captured precisely by the demand-side instrument already in the corpus: the failure of a DBA-COMM facility class is, for every dependent sector, the onset of the CDB-1 degraded-communications baseline. The forward-direction elevation obligation applies where a communications asset is the sole service path for a facility carrying a Tier-2 or Tier-3 designation in another DBA series — a 911 center, a hospital, a grid control center, a pipeline SCADA link — and concentrates, per Section 3.3, at the access boundary. For a 911 service, a sole path is counted by the transport network it rides, not by the number of circuits or routes: its paths are diverse only when they ride different transport networks (Section 3.7; *CenturyLink Outage* F-5).

6.2 Inward View — the Power Dependency

The reverse dependency runs first to the electric sector. Communications infrastructure is a continuous electric load with limited ride-through — central-office backup measured in hours

and cell-site batteries in a few hours — and its sustained operation depends on grid supply and on fuel resupply for backup generation. The coupling is bidirectional and self-reinforcing: the electric sector depends on communications for its own SCADA and control coordination, so a compound event that degrades both removes each sector's mitigation for the other. This is the same bidirectional structure the energy series treats for the gas-electric cascade, and it must be stated as a design basis condition in both directions. The two sectors' links can also be lost together. A seabed corridor shared by telecommunications cables and an energy link is lost to one anchor drag for both sectors at once, and the energy link's repair is counted in months. In the Baltic Sea cable incidents (DBA-COMM-RE-1), one ship dragging its anchor is associated, on the investigators' accounts, with damage to the Balticconnector gas pipeline and three telecommunications cables in October 2023, and another with damage to the Estlink 2 power cable and four telecommunications cables in December 2024; the cables were repaired in days to weeks, and the pipeline and the power cable were out of service for about six months each. The shared corridor is shared in exposure but not in repair. Dependents whose transport paths and energy links share such a corridor count it as one place for both, and how the energy series counts it is that series' own decision (*Baltic Sea Cable Incidents* F-3; *Transport and Backbone* §8.3, §10.7). No finding of attribution or intent is made here.

6.3 Critical-Services Elevation at the Access Boundary

The critical-services overlay is where the elevation rule fires most often. A 911 / PSAP connection, a public-safety broadband circuit, or the alerting path for the Emergency Alert System carries a life-safety consequence out of proportion to the size of the serving facility, so an access asset that is the sole path for such a service inherits a tightened acceptance criterion regardless of its own scale.

Section 7 — Military-Action Extension Applicability

[Skeleton.] Communications is, by the corpus's own analysis, among the first systems an adversary targets under military action, so the sector-level presumption toward MA applicability is strong. Application remains a per-facility determination under the four-filter applicability gate of DBA-MA-AF (MA Applicability Framework) §2 and §3, in which any one filter satisfied brings a facility into scope. For communications facilities the strategic weight is driven substantially by the outward-service dependencies of Section 6. Landing stations, internet exchange points and the timing infrastructure are the highest-value nodes.

Section 8 — Facility Class Register and Inheritance Model

8.1 Inheritance Model

Facility-class instruments of every type in the DBA-COMM series inherit the threat taxonomy, bounding methodology, consequence-analysis structure, adequacy standards, and Bright Line / Command-Broker architecture established in this framework. Each instrument adds what is

specific to its function — its bounding events, consequence chains, design parameters, and recovery architecture. The communications design basis is not part of that inheritance. Each instrument cites DBA-MA-CDB (*Military Action Communications Design Basis*) directly and states how CDB-1 through CDB-7 apply to its own control and coordination communications, as *Military Action Communications Design Basis* §3 requires of every facility class in every sector (Section 1.5).

8.2 Facility Class Register

The following table is the current facility-class register for the DBA-COMM series. All five classes are at working-draft status, issued in October 2026; the critical-services overlay is stated as an elevation overlay (Section 3.7) and has no instrument of its own. Designators were ratified 2026-08-09.

Designator	Function class (function-primary)	Governing characteristic	Status	Primary regulatory / coordination anchor
DBA-COMM-TR-FC1	Transport / backbone — long-haul & metro fiber, subsea cable & landing stations, core optical/IP backbone, satellite backbone	concentration: cable landing stations, long-haul chokepoints, IXP-adjacent routes	Working draft	FCC; Team Telecom / DOJ (subsea, foreign carriers)
DBA-COMM-AC-FC2	Access / edge — cellular RAN & 5G (towers, base stations), wireline & fixed-wireless last mile, satellite access, over-the-air broadcast (TV / FM / AM; EAS carrier)	power-dependency: CO backup 8–24 h, cell batteries 4–8 h	Working draft	FCC; state PUC; FirstNet (public-safety LTE)
DBA-COMM-CR-FC3	Core — switching, routing & signaling: mobile switching centers, internet exchange points, SS7/Diameter, BGP routing, number-portability databases	the network control plane; software-data dominated	Working draft	FCC; voluntary (CSRIC)
DBA-COMM-PN-FC4	PNT reception, timing & synchronization distribution — ground-segment reception, holdover & timing distribution the network and other sectors depend on; GPS space segment a boundary object (Space Force / GF / DIB)	cross-sector timing key; common-cause dependency across comms, grid & finance (elevated from overlay to its own class)	Working draft	EO 13905 (PNT resilience); no single regulator
DBA-COMM-OpsCo	Operations & sector coordination (coordination class) — NOCs, provisioning, and the sector coordination function	n/a (enterprise + sector commons)	Working draft	CISA (SSA); FCC; NCC / Comms ISAC; NSTAC
DBA-COMM-CS†	Critical-services overlay — 911 / NG911 / PSAP, EAS / WEA alerting, GETS / WPS priority services	elevation-driven (life-safety)	Stated (elevation overlay; Section 3.7)	FCC; state 911 authorities; CISA (priority services)

Designators were ratified 2026-08-09 and confer no completed instrument. Facility-class numbers are sequential across the series. TR-FC1, AC-FC2, CR-FC3 and PN-FC4 are physical classes; OpsCo is the coordination class and carries no FC number, mirroring DBA-ES-CntlCo. CS is a cross-cutting elevation overlay, on the author's decision of 6 October 2026 (Section 3.7).

Section 9 — Development Status and Open Items

This is a working draft of the DBA-COMM sector framework. It establishes the standalone-series placement, the supply/demand dual with DBA-MA-CDB, the function-primary taxonomy (transport, access, core, and positioning, navigation and timing as physical classes, and the OpsCo coordination class) with the critical-services overlay, the hybrid coordination-and-regulatory terrain, and the cross-sector elevation structure including the bidirectional power dependency. The design-basis envelope (Section 4), the dual-DB detail (Section 5), and the MA applicability section (Section 7) are stubs to be developed. Open items at this version: DBA-COMM-TR-FC1 (*Transport and Backbone*), DBA-COMM-AC-FC2 (*Access and Edge*), DBA-COMM-CR-FC3 (*Core Switching, Routing and Signaling*), DBA-COMM-PN-FC4 (*Positioning, Navigation and Timing*) and DBA-COMM-OpsCo (*Operations and Coordination*) are issued, completing the series' five classes; take in, at a conformance pass of this framework, the elements that the classes' reference events return to it and that the classes record for it (*Access and Edge* §10.9; *Positioning, Navigation and Timing* §11.5), among them candidate bounding events for the loss of satellite timing and for a fleet of user terminals disabled through its management system; fold the use-case material the reference-event instruments offered into the existing use cases at their next revisions, as the author decided on 10 October 2026; have the other series cite *Positioning, Navigation and Timing* §7 for their timing receivers; and read two of the communications design-basis requirements for this sector. Whether the critical-services overlay becomes a facility class was settled on 6 October 2026: it stays an elevation overlay (Section 3.7). CDB-2 and CDB-5 are still worded for the electric sector — CDB-2 names a grid's supervisory control and energy-management systems, and CDB-5 the gas–electric boundary — so until the *Military Action Communications Design Basis* words them for every sector, each DBA-COMM class reads them for its own network-management and coordination links. The reconciliation of the communications design basis with this framework, carried here as an open item through v0.4, was made at v0.5 (Sections 1.5 and 8.1).

9.1 Conferred Reference Events

On the author's decision of 6 October 2026, this framework confers a reference-event designator on each of the fifteen events its five classes proposed. A designator commits the series to two things: a row in the corpus-wide DBA Reference Event Register, and a standalone reference-event instrument for that event on the model of the water series' DBA-W-RE-1 (*Minnesota Water Campaign*) and DBA-W-RE-2 (*Oldsmar Water Treatment*). All fifteen instruments are written. Under the determination of 9 October 2026 (*Reference Events Move to Their Own Project*), reference-event instruments are held by Project 11, DBA Reference Events; this framework keeps the conferral, and the fourteen live communications instruments are held there. On the author's triage ruling of 9 October 2026 one conferral, DBA-COMM-RE-15, is recast and its designator retired (below). The designators are numbered in the order the proposing classes were issued and, within each class, in the order that class lists its events. The number records conferral, not issue, and the instruments may be written in any order. Each designator is sector-

primary and single: an event whose consequence reached other sectors is conferred here, in the sector that proposed it, and its reach is recorded in the register's impacted-sector column rather than in a second designator.

From *Transport and Backbone*, three events are conferred: DBA-COMM-RE-1, the Baltic Sea cable incidents of 2023 and 2024; DBA-COMM-RE-2, the Red Sea cable cuts of February 2024; and DBA-COMM-RE-3, the Tonga isolation of January 2022.

From *Access and Edge*, three events are conferred: DBA-COMM-RE-4, the Nashville bombing of December 2020; DBA-COMM-RE-5, Hurricane Maria in September 2017; and DBA-COMM-RE-6, the KA-SAT attack of February 2022.

From *Core Switching, Routing and Signaling*, three events are conferred: DBA-COMM-RE-7, the AT&T outage of February 2024; DBA-COMM-RE-8, the Rogers outage of July 2022; and DBA-COMM-RE-9, the Kyivstar attack of December 2023. The Salt Typhoon intrusion, which that class cites as an illustration, is conferred once, under the coordination class.

From *Positioning, Navigation and Timing*, three events are conferred: DBA-COMM-RE-10, the GPS timing error of January 2016; DBA-COMM-RE-11, the Galileo outage of July 2019; and DBA-COMM-RE-12, the satellite-navigation interference in the Baltic region from 2023. DBA-COMM-RE-12 and DBA-COMM-RE-1 are different events in the same sea, the one radio interference with satellite navigation and the other the physical severance of cables, and they are titled so that neither is read for the other.

From *Operations and Coordination*, three events are conferred: DBA-COMM-RE-13, the CenturyLink outage of December 2018; DBA-COMM-RE-14, the Salt Typhoon intrusion disclosed in 2024; and DBA-COMM-RE-15, national roaming in Ukraine from March 2022 to February 2026.

DBA-COMM-RE-15, national roaming in Ukraine, is recast on the author's triage ruling of 9 October 2026 and his ruling of 10 October 2026, because it is not a reference event under the backward test: it is withdrawn from this conferral and its designator retired. Its F-2 is merged into DBA-COMM-RE-9 (*Kyivstar Attack*) F-5 and its F-3 received there as F-9, and *Operations and Coordination* cites those findings in their place. The remainder is an illustration of the class's third reference configuration, mutual aid among competitors, at *Operations and Coordination* v0.5 §1.3, keeping the substance of its F-4 and F-6, and that class states at its §6.4 the author's acceptance of its R-4. The instrument stays with Project 9, in its archive, and does not move to Project 11.

Every class states its events as conferred and written: *Transport and Backbone*, *Core Switching, Routing and Signaling* and *Operations and Coordination* at their earlier revisions, and *Access and Edge* at v0.4 and *Positioning, Navigation and Timing* at v0.2, each of which took in on 10 October 2026 the corrections and recommendations its three events returned. *Positioning,*

Navigation and Timing draws at its §7.3 the line between the timing function of a receiver and the navigation function aboard aircraft and ships, for the author's ruling.

References

[To be completed.] Governing and inherited instruments: FD-BL (The Bright Line) and its determinations FD-BL-D1 (Mode of Discharge) and FD-BL-D2 (Indication Integrity); FD-LD (Layer Decomposition and Determinism); FD-LS (Local Sovereignty); FD-BR (The Broker); the DBA-MA parent framework; DBA-MA-CDB (Military Action Communications Design Basis), the communications design basis, CDB-1 through CDB-7; DBA-MA-AF (MA Applicability Framework); DBA Series Gaps and Recommendations (§1.7, MA applicability); the DBA-DC series (compute-substrate dual); DBA-W-RE-1 (Minnesota Water Campaign) and DBA-W-RE-2 (Oldsmar Water Treatment), the models for the reference-event instruments of Section 9.1. The Foundational Definitions and the DBA-MA instruments are cited without a version token; FD-REG (Governing Stack Register) is authoritative for the Foundational Definitions' version state. Regulatory and coordination: FCC (NORS, DIRS, infrastructure rules); CISA (SSA, NCC, GETS/WPS); CSRIC best practices; NSTAC; ICANN (DNS, at the boundary with the Information Technology sector); FirstNet; Executive Order 13905 (PNT resilience).

Reference-event instruments cited by short title and finding number: *The Baltic Sea Cable Incidents, 2023 and 2024* (DBA-COMM-RE-1), *The Red Sea Cable Cuts, February 2024* (DBA-COMM-RE-2) and *The Tonga Isolation, January 2022* (DBA-COMM-RE-3), cited as *Baltic Sea Cable Incidents*, *Red Sea Cable Cuts* and *Tonga Isolation*; *The CenturyLink Outage, December 2018* (DBA-COMM-RE-13), cited as *CenturyLink Outage*; these instruments are held by Project 11. *National Roaming in Ukraine, 2022 to 2026* (DBA-COMM-RE-15) is recast, its designator retired, and kept by Project 9 (Section 9.1). Facility-class instruments cited by short title and section: DBA-COMM-TR-FC1 (*Transport and Backbone*), DBA-COMM-AC-FC2 (*Access and Edge*), DBA-COMM-PN-FC4 (*Positioning, Navigation and Timing*) and DBA-COMM-OpsCo (*Operations and Coordination*). Eclectic Technologies, 2026.

Revision History

0.1 August 2026 — T. Roxey. Initial skeleton. Established DBA-COMM as a standalone sector series; stated the supply/demand dual with DBA-MA-CDB (DBA-COMM failure = the CDB degraded baseline for dependent sectors; CDB applied internally to comms operations); adopted the function-primary decomposition (Transport / Access / Core / OpsCo) with Critical-Services and PNT overlays; set the hybrid regulatory-and-coordination inheritance (FCC / CISA / CSRIC / NSTAC); registered four functional classes and two overlays at Anticipated; framed the cross-sector elevation, including the bidirectional power dependency with the electric sector. Design-basis envelope, dual-DB detail, and MA applicability sections stubbed for development.

0.2 August 2026 — T. Roxey. Checked the taxonomy against the CIKR sector mindmap. Folded over-the-air broadcast (television and AM/FM and satellite radio, including the EAS carrier) into the Access class (§3.3 and register AC-FC2), closing the one gap the comparison surfaced.

Added a provenance note that the PNT / timing overlay is elevated to first-class status here although source taxonomies list timing only as a data service (§3.6).

0.3 August 2026 — T. Roxey. Designators ratified; PNT elevated from overlay to its own class (PN → DBA-COMM-PN-FC4); DNS/PKI cut to the IT sector's CP class; Core retains SS7/BGP.

0.4 October 2026 — T. Roxey. Conformed to the October 2026 Edition of the Foundational Definitions on Project 0's work order of 3 October 2026. Section 5 states the scope of Bright Line placement by the determinism carve-out of FD-LD §4.3 for the NOC and core functions, and states that determinism decides scope and not placement (P9-11, wording). NOC decision-support output is no longer treated as an action: output that makes no change of its own is placed below the line and carries FD-BL-D2, and a change it recommends is one action with its acceptance. OI-1, the CDB inheritance, is not taken at this revision and moves to the next. Functional class is retired at every site under the author's determination of 30 September 2026, each site read for its sense: the decomposition tier reads function class, and the coordination layer reads coordination class (OI-15). No change to the design basis, the taxonomy or the register.

0.5 October 2026 — T. Roxey. OI-1 taken: the communications design basis is cited, not inherited. Sections 1.5 and 8.1 no longer make CDB-1 through CDB-7 part of what each class inherits from this framework; each class cites DBA-MA-CDB directly and states how the requirements apply to its own control and coordination communications, as that document's §3 requires. The version pin on DBA-MA-CDB is removed, and Section 9 no longer waits on the migration to the DBA-MA parent, which completed on 2 September 2026. Section 3.1 gives the classes as four physical and one coordination. Positioning, navigation and timing is no longer also an overlay: Section 3.7 and the register carry critical services as the one overlay, and DBA-COMM-PN-FC4 carries the timing elevation. Core is a physical class on the author's decision of 6 October 2026 (Section 3.4). The domain name system is placed with the Information Technology sector at Sections 2.2 and 4.1, as Section 3.4 already placed it. Section 7 applies the four-filter gate of DBA-MA-AF in place of the two-factor test stated here before. Section 9 names DBA-COMM-TR-FC1 as the first class instrument and records that CDB-2 and CDB-5 are still worded for the electric sector. References give designations with titles. Cover version and date corrected. No change to the design basis principles or the dual design basis.

0.6 October 2026 — T. Roxey. DBA-COMM-TR-FC1 (Transport and Backbone) issued and entered in the register at Section 8.2 as a working draft, the series' first facility-class instrument. Sections 3.2 and 3.4 state the boundary between transport and core on the author's decision of 6 October 2026: transport owns the fiber up to an interconnection building or internet exchange point, and the building belongs to the core class. Section 9 records the three reference events that instrument proposes, for this framework to confer.

0.7 October 2026 — T. Roxey. DBA-COMM-AC-FC2 (Access and Edge) issued and entered in the register at Section 8.2 as a working draft, the series' second facility-class instrument. Sections 3.3 and 3.4 state the boundary between access and core on the author's decision of 6 October 2026: a central office building and its backup power belong to access, and the switching and routing decisions made inside it belong to core. Section 9 records the three reference events that instrument proposes, beside the three proposed by DBA-COMM-TR-FC1, for this framework to confer.

0.8 October 2026 — T. Roxey. DBA-COMM-CR-FC3 (Core Switching, Routing and Signaling) issued and entered in the register at Section 8.2 as a working draft, the series' third facility-class instrument. Section 3.4 records the author's decision of 6 October 2026 that the building-level protection of core sites is cited from the data-center series rather than restated. Section 9 records the three reference events that instrument proposes, beside the six proposed by DBA-COMM-TR-FC1 and DBA-COMM-AC-FC2, for this framework to confer.

0.9 October 2026 — T. Roxey. DBA-COMM-PN-FC4 (Positioning, Navigation and Timing) issued and entered in the register at Section 8.2 as a working draft, completing the series' four physical classes. Section 3.5 records the author's decision of 6 October 2026 that the class holds the design basis for timing receivers in every sector, which other series cite. Section 9 records the three reference events that instrument proposes, beside the nine proposed by the other physical classes, and replaces the open item to scope the class against Executive Order 13905, which that instrument does.

0.10 October 2026 — T. Roxey. DBA-COMM-OpsCo (Operations and Coordination) issued and entered in the register at Section 8.2 as a working draft, completing the series' five classes. Section 3.6 states the author's decision of 6 October 2026 that the Command Broker for an action crossing carriers is federated, each carrier's qualified holder authorizing its own part and the commons coordinating without broker authority. Section 9 records the three reference events that instrument proposes, beside the twelve proposed by the physical classes.

0.11 October 2026 — T. Roxey. Section 9.1 confers DBA-COMM-RE-1 through DBA-COMM-RE-15 on the fifteen reference events the five classes proposed, on the author's decision of 6 October 2026, each committing the series to a register row and a standalone instrument, none yet written. They are numbered in the order the classes were issued and, within each class, in the order the class lists its events; each is conferred in this sector, which proposed it; DBA-COMM-RE-12 is titled distinctly from DBA-COMM-RE-1; and the Salt Typhoon intrusion is conferred once, under the coordination class. Section 3.6 records the author's decision of the same date that the Command Broker's independence in this sector is procedural, and that the commercially conflicted decisions named in Operations and Coordination §6.5 are recorded rather than structurally separated. Section 9 restates the open items and supplies a missing comma. The facility-class instruments conform at their next revisions. Revision-history entries are left as records.

0.12 October 2026 — T. Roxey. Section 3.7 records the author's decision of 6 October 2026 that critical services stays an elevation overlay and does not become a facility class: with 911 centers and the Next-Generation 911 network placed in DBA-ESS-AD-FC1 and alert origination in DBA-ESS-EMCo, the overlay owns no facility, Access and Edge §8.1 states its criterion, and no separate instrument is written. The register row for DBA-COMM-CS moves from Anticipated to Stated, and the register note and Section 9 say so. Revision-history entries are left as records.

0.13 October 2026 — T. Roxey. Takes in what the coordination class's and the transport class's shares of three reference events return to the framework. Section 3.6 records the author's decision of 8 October 2026 on DBA-COMM-RE-15 R-4: a binding governmental order is a governmental act outside the design basis that does not replace the carrier's holder, and the commons may not stay voluntary under military action (Operations and Coordination §10.11). Sections 3.7 and 6.1 count the dependence of a 911 service on one transport network under the elevation rule, and credit its paths as diverse only across transport networks (DBA-COMM-RE-13 R-5). Section 9.1 records that six of the fifteen instruments are written and which classes now state the events as conferred, and gives DBA-COMM-RE-15 its instrument's scope. An independent check against the instruments corrected two statements before issue: the 911 processor's service area and the share of the loss of 911, and the transport-network count, which R-5 makes for 911 service only. Revision-history entries are left as records.

0.14 October 2026 — T. Roxey. Takes in what the transport class's three reference events return to the framework. Section 3.7 counts, in the surviving service of a region with a path count of one, the payment and settlement links on which its economy depends and the coordination traffic of a disaster response (Tonga Isolation R-5). Section 4.1 names a ship disabled, abandoned or sunk by military action, and its wreck, as an initiating event of severance and a hazard to repair (Red Sea Cable Cuts R-3); states repair capacity in a war as the part of the fleet that owners will send, insurers cover and authorities admit, with about five months a point on the record for contested waters (Red Sea Cable Cuts R-6); and adopts for the sector the conditionality principle its classes state, as Transport and Backbone §10.1 asked, with the trigger that a path count of one accepted after the loss of a sole path carries a recorded basis (Tonga Isolation R-7). Section 6.2 records that a seabed corridor shared with energy is lost to one drag for both sectors, the energy repair in months (Baltic Sea Cable Incidents R-3). Section 9.1 records that nine of the fifteen instruments are written and that Access and Edge states its events as conferred. Section 9 counts the six instruments still owed. References name the reference-event and class instruments cited by short title. An independent check of the new text against the instruments raised 4 items, 4 taken before issue. Revision-history entries are left as records.

0.15 October 2026 — T. Roxey. Records the triage of the series' reference events, their move to Project 11, and the class conformance passes of 10 October 2026. Section 9.1 records all fifteen instruments as written and the fourteen live ones as held by Project 11 under the determination of 9 October 2026, the conferral staying here. On the author's triage ruling of 9 October 2026 and his ruling of 10 October 2026, it records DBA-COMM-RE-15 (*National Roaming in Ukraine*) as

recast, withdraws it and retires the designator: its F-2 and F-3 are carried by DBA-COMM-RE-9 (*Kyivstar Attack*) F-5 and F-9, and the remainder is an illustration of mutual aid among competitors in *Operations and Coordination* v0.5, which also states the author's acceptance of its R-4. Section 9.1 records that *Access and Edge*, at v0.4, and *Positioning, Navigation and Timing*, at v0.2, have taken in their events. Section 9 replaces the open item to write six instruments with the elements those classes record for this framework, to be taken in at its next conformance pass, and records that the use-case material is folded into the existing use cases at their next revisions. An independent check of the revision against the notes, the instruments and the classes raised 3 items: the recast instrument's finding still cited at Section 3.6 and in the References, both now conformed, and the pending notice to Project 11, given the same day. Project 11 is given notice of this revision. Revision-history entries are left as records.